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Characterization of the Influence of Moderate Temperatures on Rest Completion in Peach

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Abstract. **The chilling enhancement effect of moderate temperature (15°C) on rest completion in 'Redhaven' peach was verified and observed only when the moderate temperature occurred followed exposure to chilling temperatures. Chilling efficiency rose with temperatures between 0° and 8° if no moderate temperatures were interspersed. Although 0° was an inefficient chilling temperature, cycling between 0° and 15° resulted in as efficient bud rest release as 6°. The most efficient moderate temperature level was 13°, and the effect of moderate temperatures on rest completion seemed to increase when received during the latter stages of rest. Leaf and flower buds reacted similarly, although the reaction of the former was more prominent. A two-step scheme is proposed for the effect of temperature on bud rest. The first step involves the conversion from the unchilled to the chilled state by chilling temperatures. This stage can be reversed by high temperatures. The second stage is not reversible and involves the conversion by moderate temperatures of the unstable intermediate formed by step 1 to a stable material, which, when accumulated to a certain level, will result in rest completion.**

The influence of temperature on rest has been extensively studied. Low temperatures promote rest completion, whereas high temperatures interspersed with chilling temperatures will negate chilling (1, 3, 4). Recently, moderate temperatures (15°C) were found to enhance the chilling effect when cycled with 4° or 5° in a daily cycle (3, 9). This response is difficult to explain, since 15° has no chilling effect. This study was initiated to investigate the moderate temperature effect on rest in peach, and to propose a coherent explanation for this effect.

Materials and Methods

Peach semihardwood cuttings were rooted as described (2), and in one instance hardwood cuttings rooted (7) the previous winter were used. Each treatment consisted of 40 single-plant replicates, unless otherwise stated. Temperature-controlled chambers (±0.1°C) located at the Duke University Phytotron were used. In addition, Conviron E7H or Sherer chambers (±0.5°) were used for studies done in Israel and Athens, Ga. Light was maintained during the high temperature period and darkness during chilling.

Photosynthetically active radiation (PAR) was 50 $\mu mol \cdot s^{-1} \cdot m^{-2}$ unless otherwise stated. Following the various temperature treatments, the plants were moved to a lab for forcing at 22° ± 1° under continuous fluorescent light. Data were collected for 30 to 40 days (depending on the time of maximum budbreak), and percent budbreak was calculated.

Rooted peach cuttings were exposed to various constant temperatures and maintained in the chambers until 1200 hr of treatment temperature had accumulated. In a separate study, similar plants were exposed to 75 cycles in which 9°, 11°, 13°, 15°, 18°, or 21°C were maintained for 8 hr and 6° for the remaining 16 hr of the diurnal cycle. A 4° continuous treatment also was maintained for 1200 hr. In a third study, rooted peach cuttings exposed to 0°, 4°, or 6° were cycled for 16 hr with 15° for 8 hr. The treatments were continued until all plants had been exposed to the low temperature for 1200 hr. Similar plants were exposed to diurnal cycles (4° for 16 hr and 15° for 8 hr) during either the first, second, or third portions of the rest period. Following the accumulation of 1200 hr at 4°, the plants were removed to the laboratory as previously described. In Israel, rooted peach hardwood cuttings were exposed to temperature cycles in which 13°, 15°, or 17° were alternated (8 hr) with 6° (16 hr). PAR levels of 150 $\mu mol \cdot s^{-1} \cdot m^{-2}$ were provided during the high-temperature portion of the cycle. The treatments were continued until all plants were exposed to 1200 hr of 6°. A continuous 6° treatment was also maintained. Temperatures were determined by thermistors or thermocouples placed in the buds.

The statistical design was a complete randomized with 40 replications of one plant each. The data were converted by arcsin transformation and subjected to analysis of variance, SE calculations, or regression analysis. Where appropriate, the data were tested for significance by Duncan's multiple range test.

Results

Maximum budbreak occurred in plants chilled at 8°C (Fig. 1). Chilling at 0° induced very slight budbreak, and 14° was totally ineffective. Although both flower and lateral vegetative buds responded in a similar manner to the various forcing temperatures, flower buds were less responsive than vegetative buds. Data from a separate study with 'Coronet' plants indicated a significant increase in chilling efficiency on lateral vegetative buds with rise of temperature from 0° to 4° to 6°. (Fig. 2). When similar plants were chilled at the same temperatures for the same length of time but were exposed to 15° for 8 hr in a diurnal cycle with the low temperatures, differences among all chilling temperatures were nonsignificant (Fig. 2). Thus, alternating moderate temperature with chilling temperatures can increase the chilling efficiency of temperatures below 6°.

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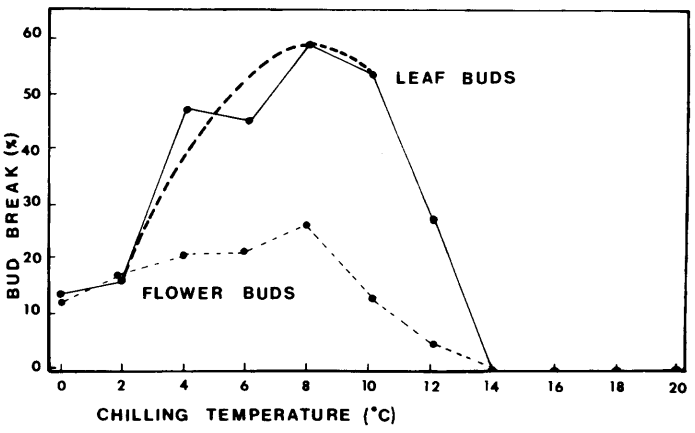


Fig. 1. Budbreak of 'Redhaven' peach plants following exposure to 1200 hr at various continuous temperatures. The response was curvilinear (Leaf buds: $y = 0.4208 + 15.4x - 1.1x^2$, $r = 0.91$; flower buds: $y = 15.4 + 2.5x - 0.26x^2$, $r = 0.81$).

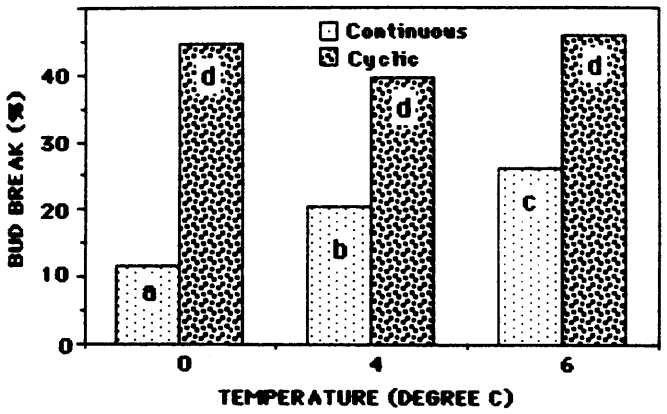


Fig. 2. The effect of exposure of 'Coronet' peach plants to 900 hr at 0°, 4°, or 6°C, applied continuously or in a diurnal cycle with 15° for 8 hr, on lateral vegetative budbreak following 28 days of forcing at 22°. Means followed by the same letter are not statistically different at the 5% level of probability according to Duncan's multiple range test.

The efficiency of moderate temperatures interspersed with chilling temperatures in a diurnal cycle was examined. Peach plants were exposed to 6°C for 16 hr followed by exposure for 8 hr to 13°, 15°, or 17°. A 6° continuous treatment served as the control. Exposure to 13° enhanced the effect of chilling on budbreak, 15° had no effect, and 17° negated chilling (Table 1). These data were obtained under relatively high light levels.

In a separate study, the effects of a wide range of moderate temperatures were examined with 'Redhaven' plants. Temperatures of 9°, 11°, 13°, 15°, 18°, or 21°C (8 hr) were cycled with 16 hr at 6° until the plants had received 1200 hr at 6°. Temperatures between 9° and 15° enhanced the chilling effect in both leaf and flower buds, 18° had no effect, and 21° negated the effect (Fig. 3). The optimum, moderate temperature appeared to be 11° for both vegetative and floral buds. However, when the data were adjusted (See Fig. 1 and Table 2) for the efficiencies of the various moderate temperatures, considering 1 hr at 6° as one unit, 13° appeared to be the most effective moderate temperature (Fig. 3).

The importance of time of exposure to the moderate temperature cycles during the rest period was examined by comparing

Table 1. The effect of continuous vs. alternate chilling on budbreak of 'Redhaven' lateral leaf buds.

Chilling treatment (°C)[z]	Budbreak (% ± SE)
No chilling	0
6	27.6 ± 6.2
6–13[y]	42.7 ± 5.2
6–15[y]	25.3 ± 6.1
6–17[y]	0

[z]One thousand two hundred chilling hours, followed by 40 days of forcing at 22°C.
[y]Sixteen hours at 6°C in the dark, 8 hr at high temperature, and 150 $\mu mol \cdot s^{-1} \cdot m^{-2}$. High temperature set according to bud temperature; 16 replications per treatment.

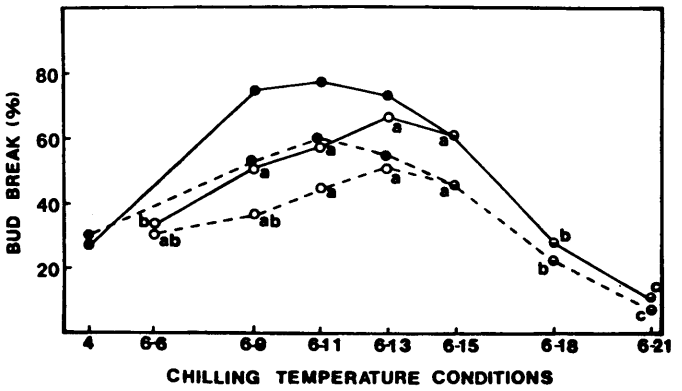


Fig. 3. The effect of exposure of 'Redhaven' peach plants to 1200 hr of continuous 4°C or 1200 hr of 6° accumulated in a diurnal cycle (16 hr at 6° and 8 hr at moderate temperature) on flower and vegetative budbreak (closed circles, and dashed and solid lines, respectively). Corrected data for equal exposure to weighted chilling according to Table 2 are represented by open circles for flower (dashed line) and vegetative (solid line) budbreak. Means followed by the same letter are not statistically different at the 5% level of probability according to Duncan's multiple range test.

Table 2. Corrected readings by equalizing exposure to chilling.

Temp regime (°C)				
Duration in cycle				Correction
16 hr	8 hr	Duration of exposure (hr)	Total weighted chilling hours (hr at 6°C = 1)[z]	factor (%)
4	4	1200	1080	+11
6	9	1800	1740	−31
6	11	1800	1620	−26
6	13	1800	1320	− 9
6	15	1800	1200	---
6	18	1800	1200	---
6	21	1800	1200	---

[z]See Fig. 1.

continuous exposure to 4°C with diurnal cycling with 15° during the first, second, or third portions of the rest period. Vegetative buds did not exhibit a response to moderate temperature in this experiment; actually, when applied during the first one-third of the chilling period, 15° inhibited vegetative budbreak. With flower buds, however, moderate temperatures exerted most of their influence when experienced during the last one-third of the rest period (Fig. 4).

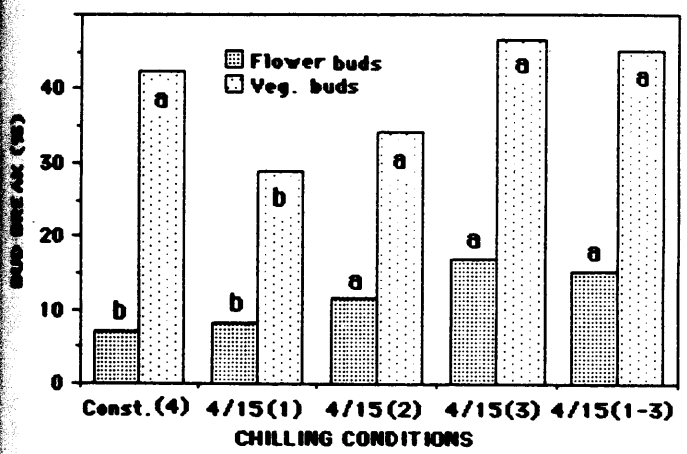


Fig. 4. The effect of timing (first, second, or last third of chilling requirement) of cyclic temperature (16 hr at 4°C and 8 hr at 15°) on breaking rest of 'Redhaven' peach buds. All buds received 1200 hr at 4°. Within bud types, means followed by the same letter are not statistically different at the 5% level of probability according to Duncan's multiple range test.

Discussion

The data obtained with 'Redhaven' peach buds clearly show that the optimum continuous temperature for satisfying the chilling requirement is 8°C. Gilreath and Buchanan (10) reported a similar temperature optimum for satisfying rest of the low-chilling 'Sungold' nectarine. Previous data (5) showed an optimum rest breaking effect at 6°. This temperature also was used in the Utah Model as the temperature optimum (12). The data presented here were obtained with better temperature control (±0.1°) conditions and a much larger number of replications than those previously reported. Temperatures ≥14° had no chilling activity and 0° had very little. Flower buds showed less differential chilling response to temperatures between 8° and 0° than did lateral vegetative buds. The variation in response to chilling among cultivars could be due to quantitative requirements or from a difference in qualitative response to low or high temperatures. The striking similarity in optimum temperature between our data, obtained with high-chilling cultivars, and that of Gilreath and Buchanan, obtained with low-chilling nectarines, suggests that optimum temperature does not differ for low- vs. high-chilling cultivars (10).

Although a constant 0°C was much less efficient than a constant 6° in satisfying rest, no difference was evident when cycled diurnally with 15°. The chilling efficiency difference between the two temperatures was lost. In all instances, the chilling efficiency of low temperatures was increased by cycling with moderate temperatures (15°), suggesting that the low efficiency of continuous low temperatures in satisfying rest is not due to a reduced chilling efficiency of these temperatures per se but to a lack of moderate temperatures. Thus, the actual chilling effect of temperatures between 0° and 8° is about the same, but the lack of moderate temperatures is responsible for the differences in chilling efficiency. Moderate temperatures were shown to promote chilling effect mostly in the latter stages of the rest period.

The chilling response to temperature cycles in which 15°C was the high temperature seemed to be modified by light level. The negative effect of light per se on chilling has been reported (6, 8). Since all chamber temperatures were set on bud rather than air temperature level, this negative effect cannot be interpreted as a temperature elevation effect. Enhancing effects on chilling in diurnal cycles with 15° reported in Figs. 2 and 3 (see also refs. 1 and 3) occurred at low PAR (50 $\mu mol \cdot s^{-1} \cdot m^{-2}$). When examined under higher PAR (150 $\mu mol \cdot s^{-1} \cdot m^{-2}$), no enhancing effect was found for 15°; 13°, however, exhibited a promotive effect.

The data presented in this report and elsewhere (1, 3, 4) allow for the presentation of a scheme to describe the temperature effects on rest in peach buds. The scheme adopted from Gregory and Purvis (11) is two-step on vernalization of rye plants. Of the two steps, the first one is reversible and the second is fixed:

$$A \underset{a}{\overset{b}{\rightleftharpoons}} B \xrightarrow{c} C$$

Where: A = the resting state;
B = the product of low temperature exposure, which can revert to "A" depending on temperature;
C = the product of B, which is fixed and thus non-reversible;
a = the chilling negation reaction (favored by high temperatures);
b = the chilling reaction (favored by low temperatures); and
c = conversion of B to C at moderate temperatures, which fixes the chilling effect.

The reactions have different temperature response curves (Fig. 5). The chilling effect (reaction b) responds to temperatures between 0° (effects at temperatures <0° are unknown) and 13°C. Chilling efficiency decreases as the temperature rises above 8°, reaching zero at 14°. Reaction a occurs at temperatures ≥16°, reaching a high level of activity at 24° (1, 3, 4). The influence of temperatures >24° is unknown. Reaction c occurs at a wide range of temperatures but is most rapid at moderate temperatures (13° to 15°). It is the most difficult to describe because its effects overlap those of reactions a and b. Reaction c occurs at 0° since budbreak will occur on plants chilled continuously at this temperature, but its maximal observed effect is at 13°. Reactions a

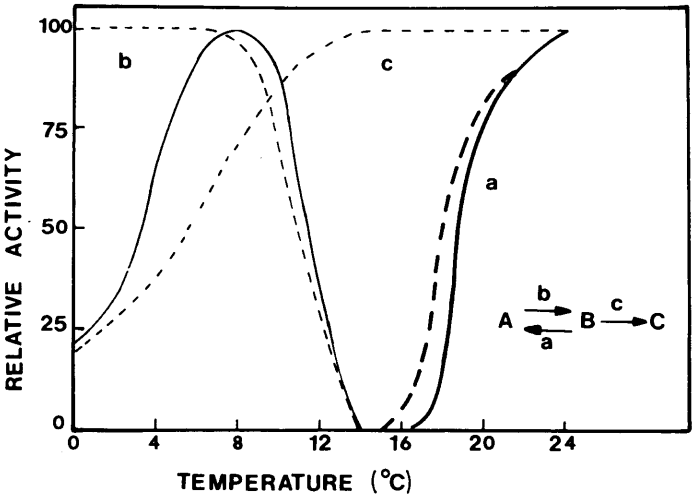


Fig. 5. Temperature response curve of the three proposed reactions occurring during chilling of resting peach buds. Curve a was obtained at the designated temperatures for 8 hr daily with 16 hr at 6°C. Solid lines represent the apparent temperature responses and the dashed lines the actual responses of the three temperature reactions.

and c compete for the same substrate B, but reaction c has a much lower Q_{10} than reaction a, which partly or fully masks it at temperatures >13°. Reaction a seems to be affected by the level and duration of the high temperature. Short diurnal periods of exposure to 20° may divide the substrate B between a and c, resulting in no negative effect (1).

This scheme can explain the promotive effect of moderate temperatures cycled with 0°C compared to chilling at 0° continuously. At 0°, reaction b can occur at optimum rates, but reaction c cannot; thus, the fully chilled state (C) is not attained to the same level as occurs when chilled at higher temperatures. However, intervening moderate temperatures allow for the rapid and efficient conversion of B to C due to optimum rates of reaction c at this temperature. Cyclical chilling of low and moderate temperatures thus results in more efficient rest completion than chilling at the most efficient continuous temperature (8°). The scheme also can explain the effect of high/low temperature cycle lengths on rest completion. Periods of high temperature (24°) applied for one-third the length of the cycle were much more effective in negating chilling in short (daily) cycles than in longer periods (4). Under long cycles, at least part of B was converted to C during the long period of chilling, whereas during the short cycles there was insufficient time for the conversion of B to C. Thus, most of the B reverted to A by reaction a, which proceeded at optimum rates. The beneficial effect of exposure to 20° following long periods of continuous chilling temperature (5) may be explained as follows. Following a long period of chilling, raising the temperature to 20° may initiate reactions a and c simultaneously, resulting in some accumulation of C. On the other hand, continuous low temperatures may be inefficient in satisfying rest because c reaction occurs too slowly.

From this scheme it is clear that any phenoclimatological model developed to simulate the response of the dormant bud to temperature must consider the effects of temperature on the three reactions proposed here.

The practical message of this study is that the level of the high (day) temperature in a diurnal cycle is of critical importance under marginal growing conditions with warm winters. This temperature either can negate or enhance, depending on its level and duration.

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